

Advanced Calculus HW

Robert O'Connell

30th January 2026

Question 1

Parameterization

To solve this problem, it may be useful to first consider a similar but easier problem, such as a circle of radius 5, lying on the x - y plane.

The parameterization of this curve would simply be

$$\vec{r}_1(t) = (5 \cos t, 5 \sin t, 0) \quad t \in (0, 2\pi)$$

since sine and cosine have periods of 2π .

Now to get the circle of radius 5 which is in the plane perpendicular to $(1, 1, 1)/\sqrt{3}$, we will construct a linear mapping, which rotates our simple circle to the desired circle.

To figure out what our mapping should be, we should figure out where the mapping should take the three standard basis vectors of $\mathbb{R}^3$.

We know the basis vector $(0, 0, 1)$ which is perpendicular to the simple circle should stay perpendicular to the circle and thus get mapped to $(1, 1, 1)/\sqrt{3}$.

To find where the other two basis vectors go, we need to find two unit vectors which lie in the plane of the desired circle which are perpendicular to each other and to $(1, 1, 1)/\sqrt{3}$.

One such unit vector $\vec{v}_1$ could be attained using the dot product

$$(1, 1, 1)/\sqrt{3} \cdot \vec{v}_1 = 0$$

$$v_{1x} + v_{1y} + v_{1z} = 0$$

(infinite such vectors exist)

$$\vec{v}_1 = (2, -1, -1)/\sqrt{6}$$

To find the final vector we use the cross product, since we need a vector perpendicular to the two other vectors

$$\begin{aligned} \vec{v}_2 &= ((2, -1, -1)/\sqrt{6}) \times ((1, 1, 1)/\sqrt{3}) \\ &= \frac{1}{3\sqrt{2}} \left(\det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \right) \\ &= (0, -3, 3)/3\sqrt{2} \\ &= (0, -1, 1)/\sqrt{2} \end{aligned}$$

We now know where this linear mapping takes the three basis vectors, so we can display this in a matrix as a linear transformation, with the images of the standard basis vectors being the columns of the matrix

$$\begin{aligned} T: \mathbb{R}^3 &\rightarrow \mathbb{R}^3 \\ T(\vec{x}) &= A\vec{x} \\ A &= \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix} \end{aligned}$$

As a final step, all we need to do is multiply our simple parameterization by our transformation matrix.

$$\vec{r}_2(t) = A \begin{bmatrix} 5 \cos t \\ 5 \sin t \\ 0 \end{bmatrix}$$

$$\vec{r}_2(t) = \begin{bmatrix} \frac{10 \cos t}{\sqrt{6}} \\ -\frac{5 \cos t}{\sqrt{6}} - \frac{5 \sin t}{\sqrt{2}} \\ -\frac{5 \cos t}{\sqrt{6}} + \frac{5 \sin t}{\sqrt{2}} \end{bmatrix} \quad t \in (0, 2\pi)$$

Which is our final parameterization of the desired circle.

Opposite Orientation

To find a parameterization of opposite orientation we need to find a bijective function which maps $t \rightarrow \tau$ such that $t = \gamma(\tau)$, for the change in parameter to change the orientation we need $\gamma'(\tau) < 0$

One such function is $\gamma(\tau) = -\tau$ with $\tau \in (-2\pi, 0)$

Recall,

$$\cos(-x) = \cos(x), \quad \sin(-x) = -\sin(x)$$

Our new parameterization is,

$$\vec{r}_3(\tau) = \begin{bmatrix} \frac{10 \cos \tau}{\sqrt{6}} \\ -\frac{5 \cos \tau}{\sqrt{6}} + \frac{5 \sin \tau}{\sqrt{2}} \\ -\frac{5 \cos \tau}{\sqrt{6}} - \frac{5 \sin \tau}{\sqrt{2}} \end{bmatrix} \quad \tau \in (-2\pi, 0)$$

Shifting the Circle

To shift the circle by a given amount in each direction say (s_x, s_y, s_z) , simply add each component to the parameterization to get

$$\vec{r}_4(t) = \begin{bmatrix} \frac{10 \cos t}{\sqrt{6}} + s_x \\ -\frac{5 \cos t}{\sqrt{6}} - \frac{5 \sin t}{\sqrt{2}} + s_y \\ -\frac{5 \cos t}{\sqrt{6}} + \frac{5 \sin t}{\sqrt{2}} + s_z \end{bmatrix} \quad t \in (0, 2\pi)$$

Question 2

$$\vec{r}(t) = (2 \cos(t^2) - 1, 2 \sin(t^2), 3)$$

- We know that both sine and cosine have periods of 2π , thus the curve starts to repeat after t^2 increases by 2π , thus we need $t^2 = 2\pi \Rightarrow t = \sqrt{2\pi}$ thus $a = \sqrt{2\pi}$
- To find the derivative of the vector valued function we simply differentiate it component-wise

$$\vec{r}'(t) = (-4t \sin(t^2), 4t \cos(t^2), 0)$$

The norm is then

$$\begin{aligned} |\vec{r}'(t)| &= \sqrt{(-4t \sin(t^2))^2 + (4t \cos(t^2))^2} \\ &= \sqrt{16t^2(\cos^2(t^2) + \sin^2(t^2))} \\ &= 4|t| \end{aligned}$$

- If $t = \gamma(\tau) = \sqrt{\tau}$, then $\vec{r}(\tau) = (2 \cos(\tau) - 1, 2 \sin(\tau), 3)$ with $\tau \in (0, 2\pi)$

$$\frac{d\vec{r}}{d\tau} = \frac{d\vec{r}}{dt} \frac{dt}{d\tau}$$

Where $\frac{d\vec{r}}{dt} = \vec{r}'(\sqrt{\tau})$ and

$$t = \sqrt{\tau} \Rightarrow \frac{dt}{d\tau} = \frac{1}{2\sqrt{\tau}}$$

Thus

$$\frac{d\vec{r}}{d\tau} = \frac{1}{2\sqrt{\tau}} \vec{r}'(\sqrt{\tau})$$

It's norm is then

$$\begin{aligned} \left| \frac{d\vec{r}}{d\tau} \right| &= \left| \frac{1}{2\sqrt{\tau}} \vec{r}'(\sqrt{\tau}) \right| \\ &= \frac{1}{2\sqrt{\tau}} (4\sqrt{\tau}) \\ &= 2 \end{aligned}$$

Question 3

$$\vec{r}(t) = \begin{cases} (a, b, c) & \text{if } t = 3 \\ (4 \sin(\ln(t+1)), -\sqrt{16-t^2}, \sinh(3t)) & \text{otherwise} \end{cases}$$

- The natural domain of a vector valued function is the intersection of the natural domains of the component functions. Remember sine and sinh can take any input, the natural log requires inputs greater than zero and square roots can not take in negatives

$$t+1 > 0, \quad 16-t^2 \geq 0, \quad t \in \mathbb{R}$$

$$t > -1, \quad |t| \leq 4, \quad t \in \mathbb{R}$$

Thus the natural domain of $\vec{r}(t)$ is $(-1, 4]$

- If a function is continuous at a point a then,

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Also, for a vector valued function then each of the limits of the components must be equal. Since for $\vec{r}(t)$ each of the components themselves are continuous then we know

$$a = x(3), \quad b = y(3), \quad c = z(3)$$

$$a = 4 \sin(\ln(4)), \quad b = -\sqrt{7}, \quad c = \sinh(9)$$

- We know in general that differentiability implies continuity while the converse of the statement is not necessarily true, but in this case, $\vec{r}(t)$ is differentiable, since all of it's component functions are differentiable at $t = 3$, the fact that $\vec{r}(t)$ is defined separately at $t = 3$ has no effect, since the curve remains continuous.

$$\vec{r}'(t) = \left(\frac{4}{t+1} \cos(\ln(t+1)), \frac{t}{\sqrt{16-t^2}}, 3 \cosh(3t) \right)$$